Yining (Carina) Fu

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Professional Summary

Biostatistics MSPH candidate with experience applying SQL, R, SAS, and Python to public health, clinical, environmental, and financial datasets. Skilled in data cleaning, statistical modeling, missing data imputation, machine learning classification, and data visualization. Experienced in analyzing ACS records and communicating quantitative findings to multidisciplinary and non-technical audiences.

Education

Emory University — Atlanta, GA
Master of Public Health in Biostatistics — *Aug. 2024 – May. 2026*

Relevant Coursework: Biostatistics, Regression Analysis, Epidemiologic Methods, Machine Learning, Clinical Trials, Statistical Computing, Categorical Data Analysis

North Carolina State University — Raleigh, NC
Bachelor of Science in Statistics; Minor in Economics — *Aug. 2019 – Dec. 2023*

Honors: Dean's Honors List, 4 semesters

Experience

SOM Clustering of Neighborhood Profiles — Atlanta, GA
Quantitative Research Assistant — *Sep. 2025 – May. 2026*

- Engineered 13 standardized variables by integrating ACS, environmental, and built-environment data to construct neighborhood profiles.
- Optimized Self-Organizing Map (SOM) using rigorous validation metrics (Gap Statistic, Elbow Method, Silhouette Analysis) to quantify cluster separation.
- Interpreted multidimensional socioeconomic patterns and assessed model classification robustness, while identifying temporal data limitations.

Chinese Academy of Sciences, Institute of Soil Science — Nanjing, China
Research Assistant — *Jan. 2024 – Jun. 2024*

- Built spatial segmentation and clustering models in SAS to identify regional soil contamination patterns.
- Conducted data normalization, quality control, and exploratory analysis to prepare environmental datasets for statistical modeling.
- Supported quantitative interpretation of soil contamination trends for regional environmental assessment.

Shanxi Securities Research Institute — Shanghai, China
Industry Intern — *Jul. 2022 – Aug. 2022*

- Integrated financial datasets from Wind and Choice to evaluate company performance and support industry research.
- Automated valuation metric calculations using Python and R, improving the efficiency of recurring analytical workflows.
- Prepared Excel-based analytical reports and presentations to summarize business insights for senior stakeholders.

Academic Projects

Aedes-Borne Viruses (ABV) Transmission Analysis Project — R

- Analyzed the spatial-temporal transmission dynamics of a 2023 ABV outbreak using test-negative design data from a cluster-randomized trial.
- Developed custom R algorithms to calculate the tau clustering statistic, evaluating localized neighborhood risk and vector control (TIRS) effectiveness.
- Executed stratified seasonal and sensitivity analyses, visualized via ggplot2, revealing ABV as a highly concentrated outbreak independent of general febrile trends.

FAERS Machine Learning Project — R

- Cleaned and merged seven FDA Adverse Event Reporting System datasets to build an analysis-ready dataset for machine learning applications.

- Developed and compared LDA, QDA, and KNN classification models to predict adverse event outcomes, achieving 80% prediction accuracy.
- Evaluated model performance and conducted gap analysis to identify limitations and potential areas for improvement.

Clinical Trials Case Study SAS

- Used SAS PROC SQL to clean and manipulate clinical trial datasets for efficacy analysis.
- Analyzed patient-level characteristics associated with clinical outcomes and treatment effectiveness.

Prospective Evaluation of Radial Keratotomy (PERK) Group Project R / SAS

- Conducted t-tests and ANOVA to assess relationships between patient characteristics and surgical outcomes.
- Produced numerical and graphical summaries to report post-surgical outcome improvements and procedure effectiveness.

Skills

Programming & Data Tools: R, Python, SAS, SQL, Java

Statistical Methods: Regression Analysis, Hypothesis Testing, ANOVA, t-tests, Missing Data Imputation, Classification, Clustering, A/B Testing

Packages & Platforms: Pandas, NumPy, ggplot2, dplyr, SAS Macro, SAS PROC SQL, Tableau, Microsoft Excel, PowerPoint

Languages: Mandarin Chinese, English